

Predicting Wildfires and their Severity Using Convolutional Neural Networks and Multi-Layer Perceptron

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Introduction

Wildfires pose an increasing threat to ecosystems, human populations, and infrastructure, particularly in regions prone to extreme heat and drought. The ability to predict wildfires and estimate their severity is important for efficient resource allocation and risk mitigation.

Traditional wildfire prediction models rely on meteorological data, historical fire records, and vegetation indices. However, advances in machine learning provide new opportunities to enhance the accuracy and efficiency of these predictions.

We aim to develop a dual-method ML approach to predict wildfires and their severity, leveraging convolutional neural networks (CNNs) for image-based fire detection and multilayer perceptrons (MLPs) for predictive analysis. Specifically, CNNs are applied to RGB imagery and aerial photographs of past wildfires to detect early-stage fires and assess their spatial patterns. MLPs will be used to analyze preexisting numerical data, such as temperature (C), FFMC indices, and relative humidity to estimate the magnitude of wildfire damage in terms of hectares (Canada Natural Resources). By combining these two models, we aim to build the framework for identifying fires using image data and then predicting their severities with numerical data. While these two datasets are different in location and context, they provide a complementary foundation for developing and testing the dual-model framework, with future work focusing on unifying geographic and temporal data to improve model integration.

Prior research has demonstrated the effectiveness of deep learning in wildfire prediction, leveraging both remote sensing imagery and structured environmental data. Studies have successfully trained neural networks on Moderate Resolution Imaging Spectroradiometer (MODIS) and Landsat imagery to detect active fires and assess burn severity. At the same time, structured data models, including artificial neural networks and support vector machines, have been used to forecast fire spread based on environmental factors.

One notable study by Oulad Sayad Younes and their team predicted wildfires using imagery and structured environmental data (Younes et al., 2019). Their methodology leverages ANN and SVMs to analyze data obtained from MODIS sensors, including the Normalized Difference Vegetation Index, land surface temperature, and thermal anomalies. This model achieved an impressive prediction accuracy of 98.32%. Their approach demonstrates the potential of

combining multiple data sources and advanced ML techniques for early wildfire prediction, aligning with our study's goal of enhancing prediction accuracy by integrating CNNs for image-based detection and MLPs for structured data analysis (Younes et al., 2019).

A study by Qassim University in Saudi Arabia used a newly available dataset to develop a more advanced ML approach (Zaidi, 2023). They were able to use PCA to reduce variables while retaining 96.65% of the variance, then trained an artificial neural network with two hidden layers, achieving a high accuracy of ~0.967 and an F1-score of ~0.971 (Zaidi, 2023).

Comparisons with traditional classifiers confirmed its effectiveness, while SHAP analysis identified key predictors like relative humidity, FFMC indices, and temperature in Celsius (Zaidi, 2023).

Claudia F. Cáceres from St. Mary's University employed GIS and remote sensing to describe forest fire risk zones in the Yeguaré region in Honduras (Cáceres, 2011). Utilizing data from NASA's Fire Information for Resource Management System spanning 2003 to 2009, Cáceres statistically identifies significant fire hotspots. To assess their contributions to fire risk, the study integrates various factors influencing fire occurrences, such as land cover, slope, aspect, proximity to roads, and population density (Cáceres, 2011). The findings indicate that approximately 30% of the study area falls within high to very high-risk zones, underscoring the efficacy of GIS-based models in forest fire risk assessment (Cáceres, 2011).

Aditya A. Deshmukh et al. from G H Raison College of Engineering utilized remote sensing data and machine learning techniques to develop a wildfire susceptibility map for California (Deshmukh et al., 2023). They integrate various environmental variables, including vegetation indices, topography, and climatic factors, into a machine-learning model to predict areas at high risk of wildfires (Deshmukh et al., 2023). The study demonstrates the effectiveness of combining remote sensing data with machine learning algorithms in identifying potential wildfire-prone regions, providing valuable insights for wildfire management and mitigation strategies (Deshmukh et al., 2023).

By leveraging multiple methodologies, wildfire prediction can be made more robust and adaptable to different ecological contexts, reinforcing the need for a comprehensive, multi-model approach.

Data Collection and Methodology

Data Collection

For this study, data was collected with the objective of wildfire prediction in mind. The Wildfire Dataset provides a valuable resource for deep learning-based wildfire detection, with 2,700 RGB aerial and ground-based imagery sourced from government databases, online platforms, and public domain repositories (El. Madafri, 2024). RGB imagery can serve as a crucial input for convolutional neural networks for image recognition, enabling the model to identify fire patterns based on spatial and textural features.

To enhance model performance, images in this dataset exhibit high resolution and scale variability, ranging from 153×206 pixels to $19,699 \times 8,974$ pixels (El. Madafri, 2024).

Preprocessing steps, including contrast enhancement, normalization, and data augmentation like brightness adjustments or edge sharpening, help improve the robustness of CNN-based models (El. Madafri, 2024). The dataset's multi-class confounding elements refine fire detection by reducing false positives, particularly in cases where clouds, sunsets, or other bright objects might be mistaken for fire (Vashan, 2021).

By leveraging high-resolution RGB imagery, we can develop and fine-tune CNN architectures to improve wildfire detection accuracy, providing critical insights for early fire warning systems and wildfire management strategies (El. Madafri, 2024). This dataset was cleaned, standardized, and scaled with Python to ensure compatibility with machine-learning models.

Another dataset provides structured environmental data related to forest fires, capturing key factors influencing fire occurrence and spread (Cortez et al., 2007; Vashan, 2021). This dataset of 517 records, sourced initially from the UCI repository, has been refined for research applications (Cortez et al., 2007; Vashan, 2021). It includes FWI index variables such as wind speed, rainfall, temperature, and humidity, which play a crucial role in wildfire behavior (Canada Natural Resources). By analyzing these factors, machine learning models can be trained to predict areas at high risk, enabling proactive wildfire management. It can serve as a foundation for models like MLPs which can analyze numerical data to detect patterns and assess fire susceptibility.

Methodology

The CNN was built using TensorFlow Keras to classify 32x32 RGB images into two categories—fire vs. no fire (Tensorflow). The model uses a sequential design, starting with two convolutional layers of 32 and 64 filters. The 32-filter layer detects visual features like edges and textures. The 64-filter layer enables the model to detect more detailed, abstract problems. “Same” padding preserves image size, and max pooling reduces dimensionality and overfitting. Batch normalization improves training stability and performance. The feature maps are flattened and passed through a dropout layer with a rate of around 0.4 to reduce overfitting, followed by a dense layer with 64 units (Tensorflow). The final layer contains two neurons with a sigmoid activation function, producing probability values for each class (Tensorflow). The model selects the class with the higher probability, making it well-suited for binary classification tasks such as fire v. no-fire scenarios. Binary cross-entropy was used to calculate the loss as it is well-suited for binary classification tasks like distinguishing between fire and no-fire images (Jshenoy, 2024). The dataset was split into 1,887 training samples and 812 validation samples, with labels encoded using one-hot vectors.

The MLP model used is a feedforward neural network to predict forest fire area in hectares from a variety of environmental and temporal features. The model takes in preprocessed input features such as temperature, humidity, wind speed, and categorical factors like month and day and passes them through two hidden layers using Leaky ReLU activation functions. L2 regularization and batch normalization were introduced to help improve generalization and stabilize training. The model was trained using backpropagation. These gradients are then used to update the weights and biases via gradient descent, which minimizes the prediction error. Five-fold cross-validation was used to ensure the model performs well across different data splits. The final ReLU output layer produces a single continuous value representing the predicted area in hectares affected by fire. We chose this output layer due to its ability to handle negative prediction values by mapping them to 0. The main performance metrics used for evaluating MLP are mean squared error (MSE) and R^2 . In addition, mean absolute error (MAE) and root mean squared error (RMSE) are also used to assess the final output (Botchkarev, 2019). The data was split in an 80% - 20% train-test format.

ML Results and Interpretation:

To determine whether or not these models will be good to use in a multi-model implementation to predict wildfires and their severity, we must first assess their performance metrics.

Convolutional Neural Network:

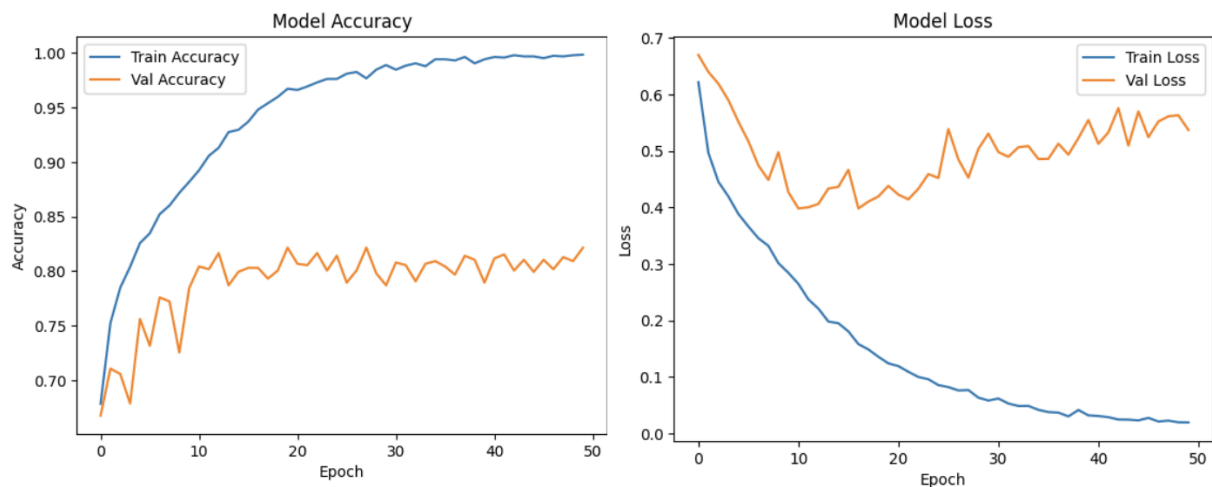


Figure 1. Model accuracy and model loss values are plotted against the validation accuracy and validation loss.

The training and validation accuracy and loss plots provide insights into the CNN model's learning behavior. The training accuracy steadily increases and approaches 1.0, while the training loss decreases consistently, indicating that the model is learning effectively from the training data (Figure 1). However, the validation accuracy plateaus around 0.82, and the validation loss begins to increase after a certain point, suggesting that the model is overfitting to the training data (Figure 1). To reduce overfitting, several regularization techniques were incorporated: a dropout layer with a rate of 0.4 to randomly deactivate neurons during training, and a dense layer with 64 units to refine the features before classification.

Despite these efforts to prevent overfitting, such as dropout, batch normalization, and dimensionality reduction via pooling, the model still overfits the training data. This can be because the dataset is imbalanced, limited in size, or lacks variability. Techniques like data augmentation such as rotating, flipping, cropping images, or using a simpler architecture can be introduced to further reduce overfitting.

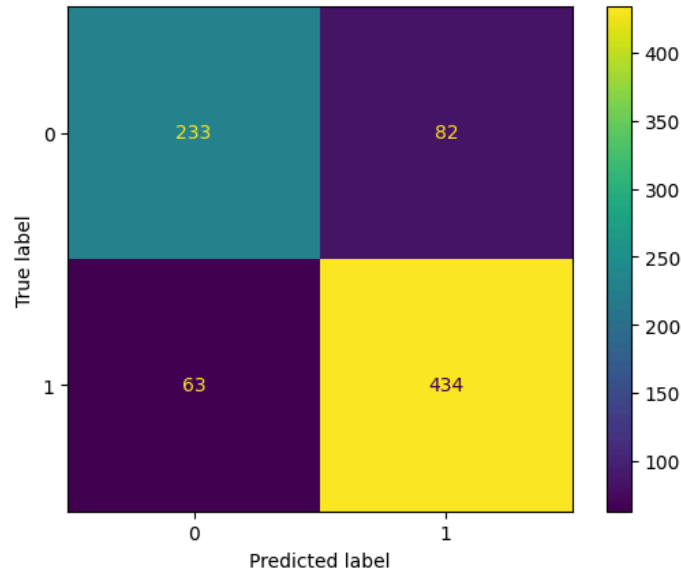


Figure 2. Confusion matrix of CNN image model.

The confusion matrix shows the classification results of the CNN model with 233 true negatives, 434 true positives, 82 false positives, and 63 false negatives of fire v. no-fire (Figure 2). This indicates that the model performs well in correctly identifying both classes, with a relatively low number of misclassifications (Figure 2). However, the false positive and false negative counts suggest there is still room for improvement in fine-tuning the model to reduce these errors.

The final evaluation metrics reinforce these observations. A training accuracy of 0.7877 and a loss of 0.6182 indicate that the model performs reasonably well on the training set. The validation loss of 0.5373 and validation accuracy of 0.8214 demonstrate that the model generalizes moderately well to the validation set. However, the gap between training and validation performance suggests that even more regularization techniques could be applied to mitigate overfitting and improve generalization.

The CNN model achieves decent performance, particularly with a validation accuracy of 82.14%, and there is potential for further optimization. Addressing overfitting and refining the model's architecture or hyperparameters could lead to improved performance on both the training and validation datasets. With the decent performance, we have confidence that this model can be implemented in a multi-model framework for predicting forest fires and their severity.

Multilayer Perceptron:

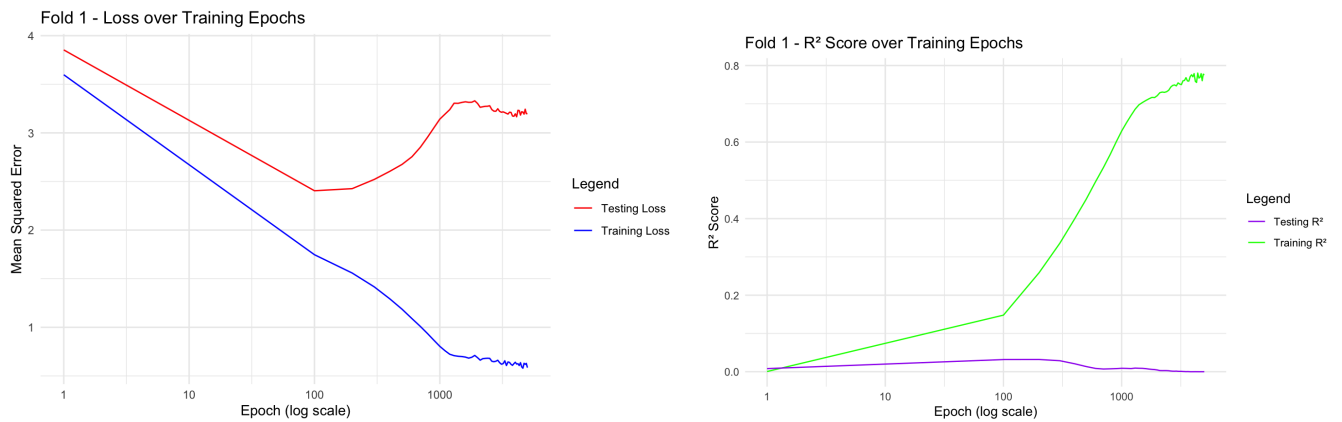


Figure 3. Loss over epochs (log scale) of training and testing data for MLP model in fold 1.

The MLP model was trained using a five-fold cross-validation framework to evaluate its ability to predict a continuous outcome. Across all folds, the training loss consistently decreased over time, demonstrating that the model was able to learn from the training data. However, the testing loss plateaued early and eventually increased in most folds, indicating that the model overfitted to the training set (Figure 3). This pattern is visible in the loss curves, where the gap between training and testing loss grows wider as training progresses (Figure 3). R^2 plots show a similar trend: while the training R^2 steadily improves and approaches values above 0.7, the testing R^2 remains close to zero throughout training, suggesting the model struggles to generalize the patterns found in the data (Figure 3).

Final evaluation metrics reflect this issue. While training losses across folds were relatively low, ranging from 0.50 to 0.57, testing losses remained much higher, between 2.26 and 3.92.

Furthermore, the R^2 scores on the test sets remained close to zero, with the highest reaching only 0.036, indicating that the model fails to explain variance in the unseen data. This consistently poor performance across all folds suggests a fundamental limitation in the model's ability to generalize, pointing to a design issue rather than variability introduced by any specific data split.

The averaged evaluation metrics across the five folds provide insight into the model's performance in predicting the log-transformed forest fire area. The MAE of 2.93 indicates that, on average, the model's predictions deviate from the true transformed values by approximately

2.93 log-transformed hectares. This reflects the typical magnitude of prediction errors without considering their direction. The RMSE of 1.71 suggests that the model's predictions deviate from the true values by about 1.71 log-transformed hectares on average, with larger errors being penalized more heavily due to squaring. This shows the presence of some larger deviations in the predictions. The MSE of 8.94 represents the average squared error, which is useful for optimization during training but less interpretable due to being in squared units of the log-transformed target variable. These metrics suggest that the model struggles to make accurate predictions, with errors that are relatively large in the context of the log-transformed scale. This performance, combined with low R^2 scores, indicates that the model has a limited ability to generalize to unseen data.

The observed overfitting likely arises from several factors. The model architecture may be too complex relative to the dataset size of 517 records, allowing it to fit the training data too closely. Such a small sample restricts the model's ability to capture patterns, increasing the likelihood that it memorizes noise in the training data. In addition, the absence of regularization techniques such as dropout, early stopping, or weight decay could enable the model to continue optimizing long past the point of generalizable learning. The quality or informativeness of the input features may also be insufficient for the task, limiting the model's capacity to extract meaningful patterns.

To address these issues, several improvements can be made. Introducing additional regularization techniques such as dropout layers can help constrain the model and reduce overfitting. Early stopping based on validation loss would prevent the model from training beyond the point of generalization (Olamendy, 2024). Reducing the number of layers or parameters could also improve test performance by limiting model complexity. Feature engineering to identify and include more informative variables, or expanding the dataset to include more representative samples, may help improve the model's predictive power and overall robustness (Patel et al., 2024). The dataset only focused on a single park and could be improved with data from parks around the world. A larger, more diverse training set could reduce variance and improve the model. With a lack of performance, we do not have confidence that this model can be implemented in a multi-model framework for predicting forest fires and their severity, and adjustments must be made.

Conclusions and Future Work

This study explored the performance of two machine learning models, a convolutional neural network for image classification and a multilayer perceptron for regression. Their application was combining the two for environmental data tasks. Both models revealed useful insights into their capabilities and limitations in handling small or moderately sized datasets, highlighting areas for improvement in future implementations.

The CNN model achieved a validation accuracy of 82.14% when classifying 32×32 pixel RGB images into binary categories of fire vs. no-fire. This suggests that the model is capable of extracting meaningful spatial features from the input images and can provide reliable predictions in most cases. However, the model exhibited overfitting, as indicated by the divergence between training and validation losses and the plateauing of validation accuracy (Figure 1). While standard regularization techniques such as dropout, batch normalization, and max pooling were applied, they were insufficient to fully prevent overfitting, likely due to the dataset's limited size or variability. Future work should focus on expanding the training set through data augmentation methods like image flipping, rotation, and cropping to introduce more variation and reduce model memorization. Exploring simpler or more tailored CNN architectures could reduce model complexity and improve generalization.

In contrast, the MLP model, designed to predict a continuous outcome, showed promising training behavior but failed to generalize to unseen data. This was shown from the high training R^2 values and low training loss, alongside persistently poor testing metrics. The most influential metrics are low R^2 and high loss across all five cross-validation folds. These results suggest the MLP was overfitting due to a mismatch between the model complexity and the small dataset of 517 records. The poor R^2 values on the test set suggest that the input features themselves may not contain enough signal to support accurate prediction. For future work, implementing dropout and other regularization techniques, applying early stopping, and simplifying the network could all improve performance. Feature importance could also be used to determine which data points are most influential for a more concise model.

Both models show the importance of matching model complexity with dataset features and rigorously validating performance to avoid overfitting. More robust evaluation strategies, such as stratified cross-validation or bootstrapping, should also be considered to better assess model reliability (Lee, 2025). Increasing the dataset size and diversity through data collection or augmentation could improve the models' performance. While the CNN demonstrates promise and could be incorporated into a multi-model prediction system for quantifying forest fire severity from imagery data, the MLP model requires substantial improvements in both design and input features before it can contribute meaningfully to such a framework.

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